

TOWARD MULTISCALE LYMPH NODE MODEL: T CELL SEARCH STRATEGY STUDY

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T cell search in DCs

- Naïve T cells continuously circulate, visiting different lymph nodes (LNs), searching for their cognate antigen presented by dendritic cells (DCs)
- Fraction of naïve T cells with TCRs for a particular antigen can be very small
→ **Effective search strategy** is crucial for initiating adaptive immune response
- Not known, whether chemotaxis is employed in the search
- Not possible to reliably determine based on available data -> modelling was used to answer the question, but different groups yielded conflicting results

Our goal:

Find out which of the search strategies is the most effective based on different parameters settings

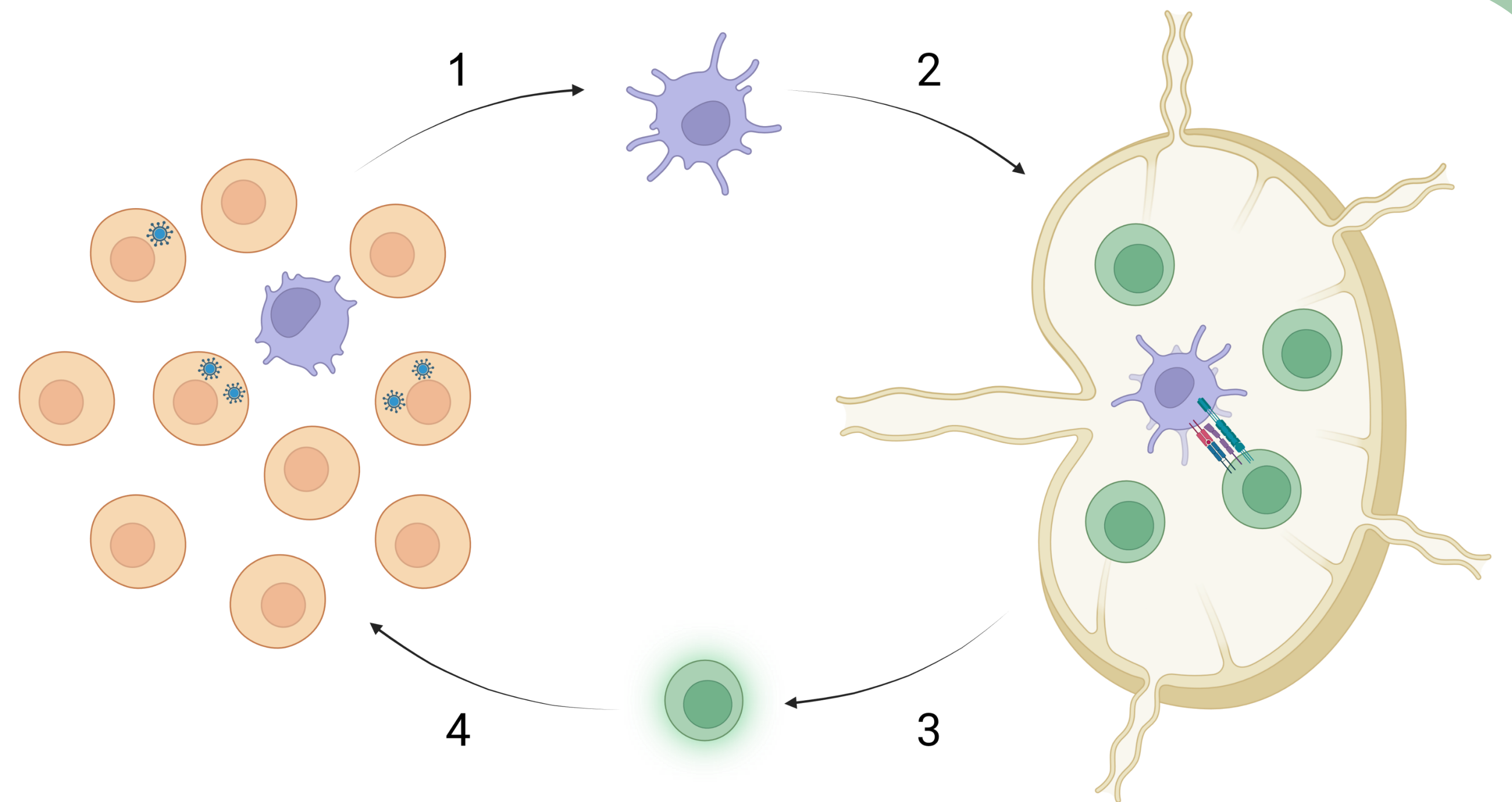
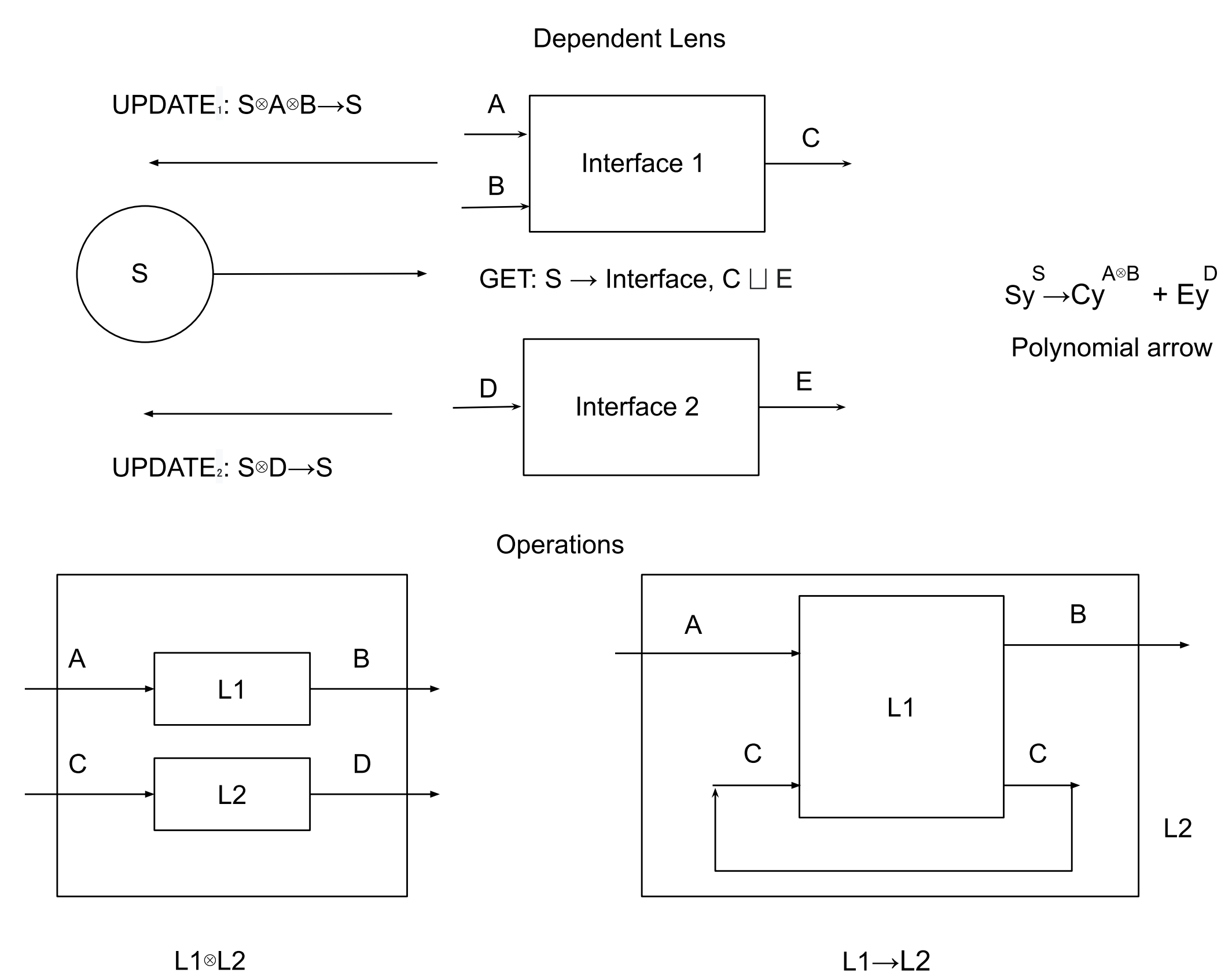


Fig 1: Interplay of DCs and T cells in immune response
1: DC maturation and migration. 2: T cell search, T cell + DC contact.
3: T cell differentiation, proliferation, and migration. 4: T cell joins immune reaction in tissue

Theoretical base of our models

Agent based models

- *Agent* = Hierarchical Dependent State Machine (Dependent Dynamical Systems), modeled as arrows (dependent lenses) and operations in the attributed category of polynomial functors (Poly). (1)
- *Interactions and composition of agents* = composition of arrows and operation in the attributed Poly.
- Agent types:
 - Environments
 - General
 - Spatially inhomogeneous
 - Reaction-Diffusion systems
 - Physical constraints
 - Cellular Potts Model
 - Entities:
 - Simple - no inner structure
 - Tokens, Chemical species (as tokens), ...
 - Complex - inner structure
 - Membranoid - inner structure with interface
 - Cells, Organs, ...



Results

Tested strategies

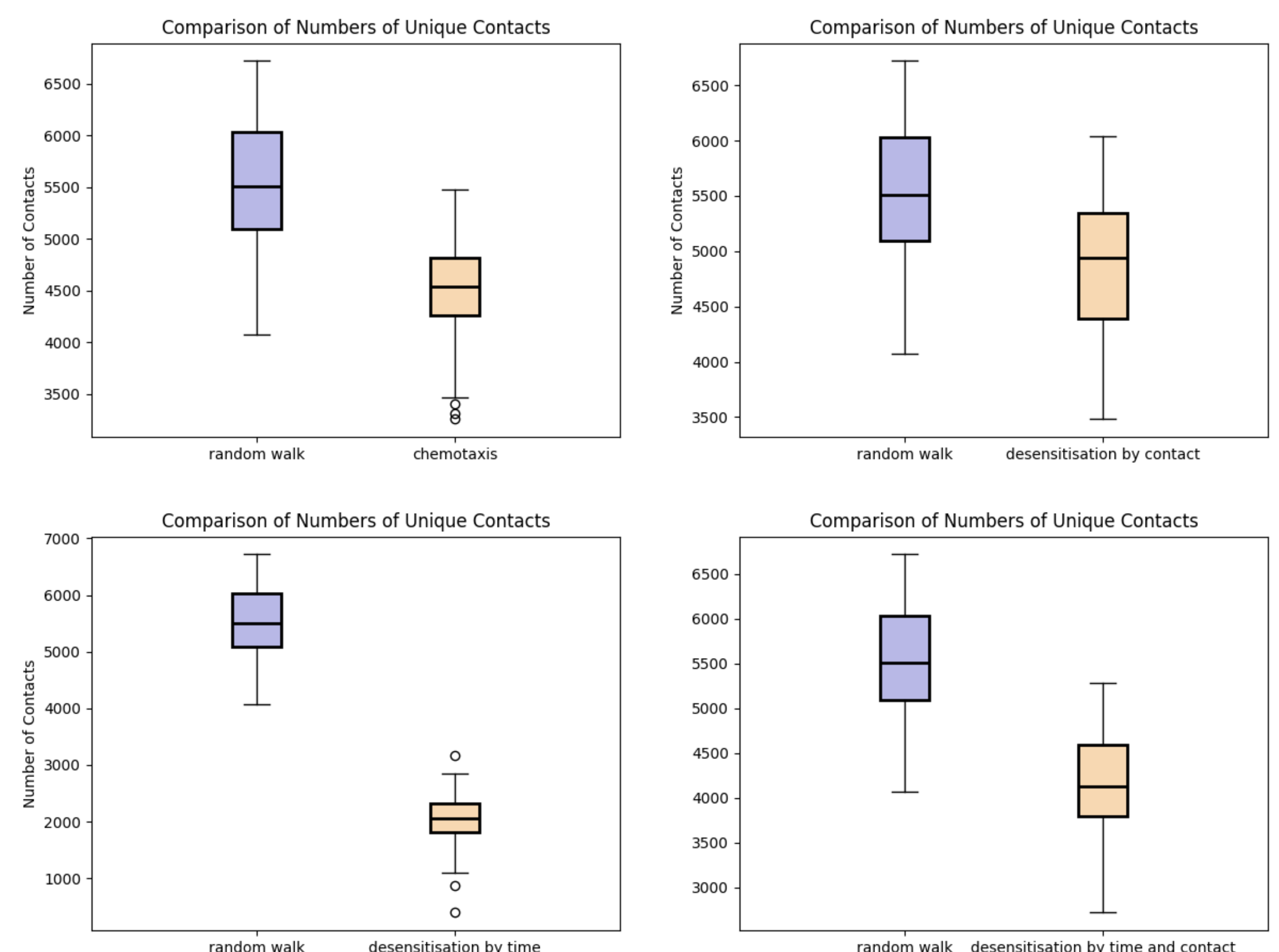
- *Random walk*
 - T cells migrate randomly
- *With chemotaxis*
- *Desensitisation by contact*
 - Insensitive to chemokine for a period of time after contact with DC
- *Desensitisation by time*
 - Insensitive to chemokine after a period of time of being in the gradient
- *Desensitisation by time and contact*
 - Both aforementioned desensitisation rules apply
- T cells resensitise after a given time interval in all desensitisation strategies

Parameters

- parameters were taken from previous works (2, 3)
- parameters in plots: 6700 T cells, 900 obstacles

Discussion

- Effectivity of the strategies assessed based on the number of unique contacts between cells
- Most effective strategy was random movement (as in 3) for current physiological setting we tried
- Some preliminary results with different parameters setting were showing that random walk might not be the most effective strategy
- Our current hypothesis for random walk as a winning strategy is that without chemotaxis, T cells could distribute more uniformly in the space and therefore all DCs could have equal chances to be contacted
 - In chemotaxis strategies, the closest DC is preferred, which might result in accumulation of T cells in some areas and therefore slow down the search



Future plans

- Further search through parameter space
- Multiscale model of LN

Find out more

- If you would like to know more, please contact the authors via strachas@natur.cuni.cz

References

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3. Vroomans, R. M. A., Marée, A. F. M., De Boer, R. J., & Beltman, J. B. (2012). Chemotactic Migration of T Cells towards Dendritic Cells Promotes the Detection of Rare Antigens. *PLoS Computational Biology*, 8(11), e1002763. <https://doi.org/10.1371/journal.pcbi.1002763>
- Fig 1 Created with [BioRender.com](https://www.biorender.com)
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